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C23-C-401

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCE – FOURTH SEMESTER EXAMINATION

CONSTRUCTION TECHNOLOGY AND VALUATION

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State any three factors influencing the strength of concrete.

2. Define the term concrete and state its ingredients.

3. State any three needs of mechanization in construction.

4. List any three types of compaction equipment.

5. State three functions of sunshade.

6. List out six causes of fire in buildings.

7. Stage any six effects of earthquake.

8. State the importance of seismic zoning.

9. Define *(a)* scrap value and *(b)* salvage value.

10. Define depreciation.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. State the properties and uses of *(a)* Fiber-reinforced concrete and *(b)* Self-compacted concrete.

12. Explain hot weather concreting.

13. Describe the construction and working of power shovel.

14. List five requirements of *(a)* good ventilation and *(b)* good lighting in buildings.

15. Draw the layout of on air-conditioning system and explain briefly.

16. What are the general principles for improving the earthquake resistance in buildings?

17. A building was constructed 20 years back on a plot of area 250 sq.m. The plinth area of building is 60 m^2. The present cost of construction of building is Rs. 7,00,000. The cost of land is Rs. 4000 per m^2. The rate of depreciation of building is 1%. Calculate the value of the property.

18. A motor vehicle was purchased for Rs. 80,000 in the year 2012. The salvage value of the vehicle is Rs. 35,000 in the year 2017. Calculate the annual depreciation by *(i)* Straight line method and *(ii)* Constant percentage method.

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